Access Remote Server (SSH)

Friday, September 12, 2025 6:21 AM

Activate SSH on Machine

1. sudo rm -f /etc/ssh/sshd_not_to_be_run
2. sudo systemctl enable ssh
3. sudo systemctl start ssh

Deactivate SSH on Machine

1. sudo systemctl stop ssh
2. sudo systemctl disable ssh

Enable/disable password authentication

1. Edit sshd_config file

   sudo vi /etc/ssh/sshd_config
2. Change *passwordauthentication* value:

   no -> yes
3. Restart SSH service

   sudo service ssh restart

Setup public/private key authentication for a user

1. Create public/private keys

   ssh-keygen
2. Share the public key (i.e. with remote server):
   a. Create the .ssh folder if not exists on remote server

   mkdir .ssh
3. Copy the public key to /home/<username>/.ssh/authorized_keys
   scp <filename> username@remote_server:./ssh/authorized_keys

   ---

   ssh-add ~/.ssh/private_key

Disable root password login -

1. sudo nano /etc/ssh/sshd_config

   PermitRootLogin no

   PasswordAuthentication no

2. Reload sshd service - sudo systemctl reload sshd
3. eval `ssh-agent -s`

Few Useful Commands

Friday, September 12, 2025 6:24 AM

For Kernel and OS version

Uname –a

lsb_release –a

cat /etc/issue.net

cat /etc/os-release

View dir file in size

ls -l --block-size=M

du -sh (for dir size)

df -h (system space)

stat -c "%a %n" dir (see permission in octal)

Process and Ports

sudo lsof -i:8080

kill -SIGKILL pid

kill -9 $(lsof -t -i:8080)

netstat -tulpn (list of open ports)

top

Detach Process from terminal

1. Start process

   sudo ls &

2. list jobs

   Jobs

3. Disown process

   disown -h %jobID

Find Utility

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Delete files older than certain time

```
find ./files/ -mtime +30 -name '*.pdf' -exec rm -rf {} \;
```

Delete all files except zip

```
find /directory/ -type f -not -name '*.zip' -delete
```

Find files between dates

```
find -newerct "1 Apr 2021" ! -newerct "16 Apr 2021" -ls | sort
```

Files should be 644

```
sudo find /var/www/html/acti-labs -type f -exec chmod 664 {} \
```

Folders should be 755

```
sudo find /var/www/html/acti-labs -type d -exec chmod 2775 {} \
```

Users & Groups

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Create user

sudo adduser <username>

Add user to sudo group

usermod -aG sudo username

Delete User

sudo userdel -r sajidali

Managing Groups

1. Create

   groupadd mygroup

2. Delete

   groupdel mygroup

3. Add user to group

   usermod -a -G sudo linuxize

Permissions

Please see lecture slides

Input / Output Redirection

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Use of >, >> & <

Redirect standard output to a file, the ">" character is used and ">>" is used to append not overwrite

ls -l > output_to_a_file.txt

Redirect standard output to a file, the ">" character is used like this

ls -l > output_to_a_file.txt

Redirect standard input from a file instead of the keyboard, the "<" character is used

sort < students_reg.txt

Can you take input from a file and after sorting save it to another file?

Your answer …

Pipelines

Helps to connect multiple commands, e.g.

Standard output of cat is redirected and now becomes input for lpr (printer)

cat students_list.txt | lpr

Apache

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sudo apt-get install apache2

Remove Apache2

sudo apt remove --purge apache2

Sudo apt remove apache2

sudo apt autoremove

Service Control

sudo systemctl stop apache2
sudo systemctl start apache2
sudo systemctl status apache2

If apache2 service is masked

sudo systemctl unmask apache2
sudo service apache2 restart

Cronjob

Friday, September 12, 2025 6:33 AM

Introduction

Cron is a utility program that lets users input commands for **scheduling tasks repeatedly** at a specific time. Tasks scheduled in cron are called cron jobs.

A cron file is a simple text file that contains c**ommands to run periodically at a specific time**. The default system cron table or crontab configuration f**ile is /etc/crontab**, located within the **crontab** directory **/etc/cron.*/**.

```
# Example of job definition:
# .---------------- minute (0 - 59)
# |  .------------- hour (0 - 23)
# |  |  .---------- day of month (1 - 31)
# |  |  |  .------- month (1 - 12) OR jan,feb,mar,apr ...
# |  |  |  |  .---- day of week (0 - 6) (Sunday=0 or 7) OR sun,mon,tue,wed,thu,fri,sat
# |  |  |  |  |
# *  *  *  *  * user-name command to be executed
17 *    * * *   root    cd / && run-parts --report /etc/cron.hourly
25 6    * * *   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.daily )
47 6    * * 7   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.weekly )
52 6    1 * *   root    test -x /usr/sbin/anacron || ( cd / && run-parts --report /etc/cron.monthly )
#
```

Important link to follow

https://crontab.guru/

See error log if needed

grep CRON /var/log/syslog